

U.S. Multi Industry & Electrical Equipment

Liquid Cooling: What does Google's Brazos mean for the broader CDU ecosystem?



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Google recently released early specifications for a liquid-to-air (L2A) CDU named "Brazos". Like project "Deschutes" that came before it in 2025, Brazos simply announces a CDU reference design for the Open Compute Project's vendor ecosystem (which counts Vertiv, nVent, and Boyd as members) to produce. **It is not a competing product to Vertiv/Boyd/nVent/Motivair's existing L2A CDU lines that Google will manufacture; hyperscalers are not in the business of making this equipment.**

Prima facie, the specifications on Brazos-class CDUs seem relatively low. At 60kW of cooling capacity, it cannot even cool one Blackwell rack (which requires ~120kW of cooling capacity). Interestingly, Brazos-class CDUs are designed to draw DC power (vs. many other L2A CDUs that are designed for AC power instead).

We believe these CDUs are designed to be OCP compliant, and fit into legacy hyperscaler deployments which tend to have DC power infrastructure already. The cooling capacity of 60kW does not work well for frontier training models, but can support inference at that range. Given concerns around how quickly data center capacity can get up and running, creating an inference ecosystem that can run in existing facilities with simple L2A CDU retrofits seems like a good move (especially when inference looks set to grow faster than training through 2030). **While not explicitly stated, we infer that this is the primary purpose of the Brazos project.**

While hard to quantify, we see two mid-term risks for CDU manufacturers. First, there is real commoditization risk of the inference CDU ecosystem. Brazos specs are meaningfully easier to deliver than Deschutes. While there will be a service attach, margins will be lower vs. flagship training CDUs. The deciding factor here is inference rack densities; we are unsure what this looks like for purpose-built inference silicon from hyperscalers. If it remains around 60 kW/rack, then a meaningful share could be cooled by L2A Brazos-style CDUs. If not, larger Deschutes-class CDUs will be needed instead (which we would expect have higher margin and stickier service attach).

Second, we may also see some hyperscaler demand shift from greenfield to retrofits, especially if projects get delayed. While we do not see that in stranded capacity numbers in our data center capacity tracker, there has been some anecdotal input from companies that they are seeing a few customers (both hyperscalers and colos) push out deliveries because of a lack of readiness / project delays. We're not saying this **WILL** happen, but we think having these standardized L2A CDUs creates optionality for hyperscalers if they need to accelerate retrofits to meet inference demand in the future.

We do not view this announcement by Google as imminently alarming for CDU manufacturers, especially those that focus on technical excellence at the top of the stack. Some level of CDU commoditization was always expected by the market. While we think this announcement accelerates the commoditization trend for lower spec. products, we still believe an innovation premium exists for flagship CDUs, and expect it to continue as top-tier liquid cooling technology evolves.

BERNSTEIN TICKER TABLE

Ticker	Rating	16 Jun 2026		Price Target	TTM Ret. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
VRT (Vertiv)	O	USD	299.60	416.00	132.8%	USD	4.20	6.52	9.21	71.4	45.9	32.5
NVT (nVent)	O	USD	167.34	218.00	113.2%	USD	3.34	4.79	6.19	50.0	34.9	27.0
SU.FP (Schneider)	O	EUR	276.95	310.00	7.3%	EUR	8.43	10.22	11.95	32.9	27.1	23.2
ETN (Eaton)	O	USD	407.71	534.00	(3.9)%	USD	12.07	13.29	16.32	33.8	30.7	25.0
SPX			7,511.35									
EDME			1,577.98									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **Vertiv** Outperform with a target price of \$416.We rate **nVent** Outperform with a target price of \$218.We rate **Schneider** Outperform with a target price of €310.We rate **Eaton** Outperform with a target price of \$534.

DETAILS

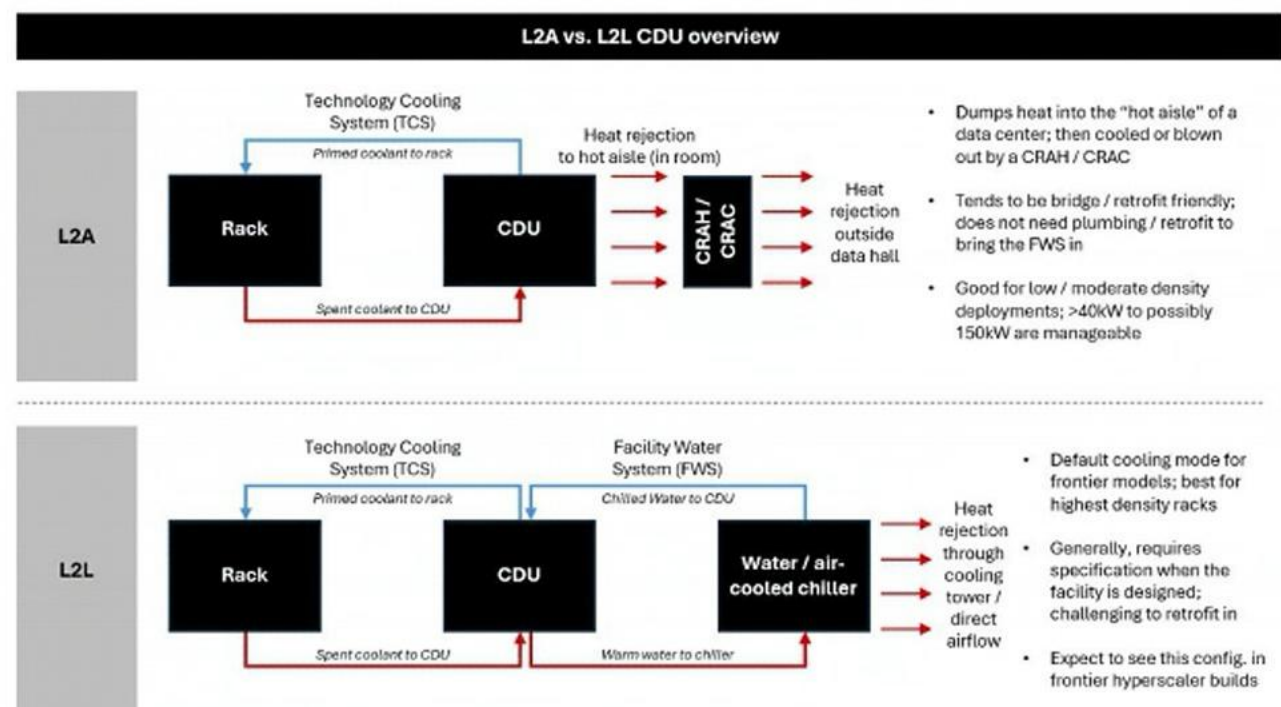
Context

Hyperscalers have historically pushed the "Open Compute Project" or OCP, a non-profit that published open-source specifications for data center hardware. Founded by Meta, it now has participation from Google, Microsoft, and Oracle (although Amazon is a notable missing name) and a host of other vendors. In 2025, Google published details of Project Deschutes as a part of the OCP; this was a set of technical specifications and standards for compliant Liquid to Liquid (L2L) CDUs (with 2MW of cooling capacity at a 3°C approach temperature). Vertiv, nVent, and Boyd all now have Deschute-compliant units available in their product catalogs. Yesterday, Google announced that they would specify another CDU design - this time for Liquid to Air (L2A CDUs) named project Brazos (in line with Google naming these projects after rivers). This note walks through implications for the broader CDU ecosystem.

L2L vs. L2A: What's the difference?

In our [CDU primer](#) earlier this week, we focused mostly comparing on flagship L2L CDUs. L2L units are named as such because they reject heat between two liquid loops (the technology cooling system and facility water system). In contrast, the L2A configuration we are talking about today only has one liquid loop (the technology cooling system) which rejects heat as air into the "hot aisle" of a data center. A CRAH or CRAC (Computer Room Air Handler or Computer Room Air Conditioner) then blows the hot air out of the hall / building. Generally, L2L cooling is preferred for all greenfield builds or situations where rack densities cross 150kW (it is much more energy efficient). In contrast, L2A builds are preferred when retrofitting an existing data center because it does not need the facility water system loop (eliminating the need for complex piping retrofits) or when cooling capacities range between 40kW to ~150kW (below which plain old - but cold - air can be used to extract heat from the chips). Both L2A and L2L are still types of liquid cooling (i.e., they need cold plates and coolant to extract heat from chips), they simply differ in terms of how they reject heat from the TCS to the outside of the data center.

EXHIBIT 1: Distinction between Liquid-to-Air and Liquid-to-Liquid CDUs

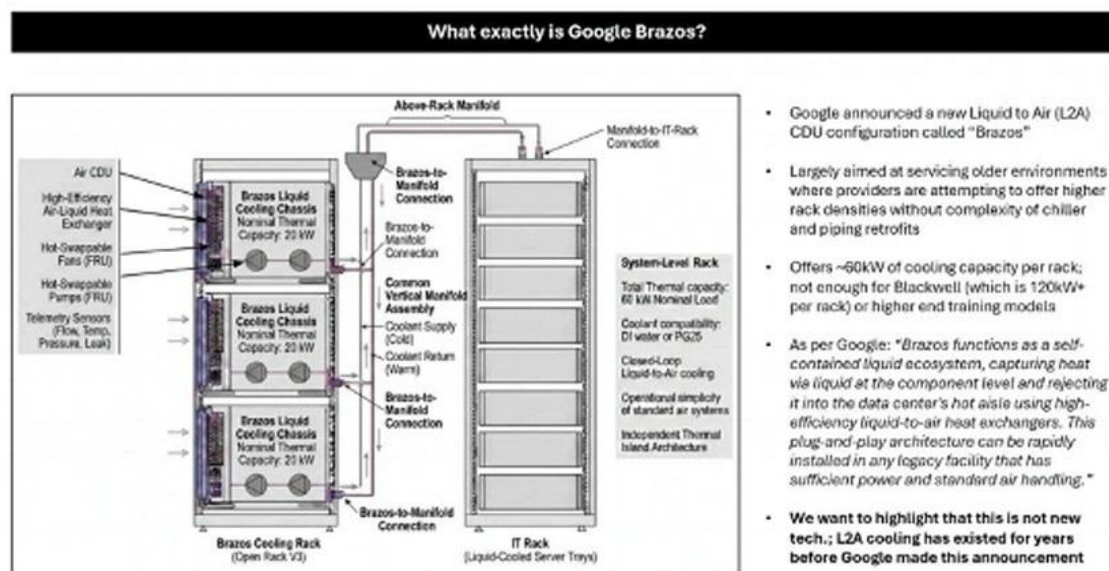


Source: Bernstein Analysis and Estimates

Thoughts on Brazos specifications

When we look at the specifications offered by Brazos, prima facie it does not look that impressive or demanding. 60kW of capacity cannot cool even a single Blackwell rack (which requires ~120kW of cooling power) - players like nVent, Motivaair, and CoolIT go far higher on their cooling power for L2A CDUs. The power feed is DC (not AC) and designed to be pulled directly from busbars which is distinct from most other L2A CDUs available in the market today which are designed for AC use.

EXHIBIT 2: Project Brazos Overview



Source: Bernstein Analysis, Company Reports (Google)

EXHIBIT 3: Brazos L2A CDU specs. seem to target inference in OCP compliant environments

L2A Product comparison of Brazos vs. peers						
Company	Google Brazos	Boyd RAA32-10U21	Vertiv CoolChip	nVent HRU	Motivaair HDU	CoolIT AHx240
Cooling Capacity	60 kW	32 kW	70 kW	100 - 120 kW	150 kW	240 kW
ATD	Not stated	24°C	11°C	Not stated	Likely ~10°C	Not stated
Power Supply	40 - 60V DC	1P AC	1P AC	1P AC	1P AC	1P AC

1 Cooling power is too low to work with Blackwell or above; seems like Brazos is more focused on inference capacity through retrofits

2 Direct DC connectivity via OCP-compliant busbar is unique to Brazos vs. other L2A CDUs; will not work in legacy colocation / enterprise builds

Source: Bernstein Analysis and Estimates, Company Reports

However, our biggest takeaway is that this design is not intended to compete with existing products. It is designed to be OCP compliant, and fit into legacy hyperscaler deployments which tend to have DC power infrastructure already and not AC. The cooling capacity of 60kW does not work well for frontier training models, but can support inference at that range. Given concerns around how quickly data center capacity can get up and running (see our data center tracker for more insight on that), creating an inference ecosystem that can run in existing facilities with simple L2A CDU retrofits seems like a smart move (especially when inference looks set to grow faster than training through 2030). **While not explicitly stated, we think this is the primary use case of the Brazos project.**

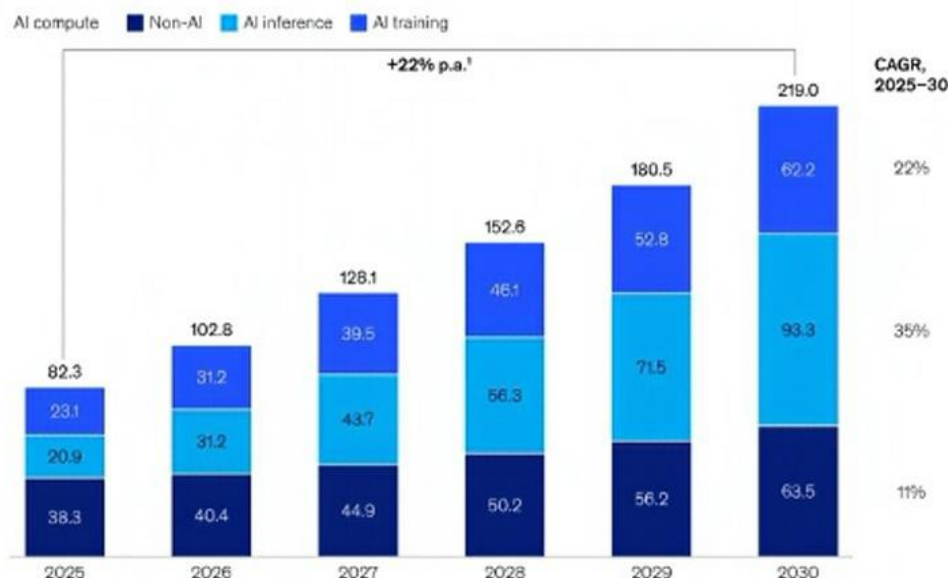
Our inferred implications on the CDU market

First, it is important to state what Brazos **is not**. It is not a CDU that Google is building; they are not in the business of making liquid cooling equipment. It is simply a set of technical specifications / requirements they will put out for their manufacturer ecosystem to build. As per Google: *"In the coming months, we will formally open-source the technical specifications, design principles, and visual assets of Brazos through industry forums. As part of a broader infrastructure portfolio that continues to leverage waterless air-cooled systems alongside liquid cooling, Brazos represents one of many innovations we are contributing to the open hardware ecosystem. We invite system architects, manufacturers, and thermal engineers to evaluate these designs to scale rack-mounted cooling infrastructure for the high-power computing demands of the future."* We also do not think this creates a structural near-term shock. CDUs will continue to see extended lead-times, and we expect L2L to be where most manufacturer focus stays.

We see two major risks for the mid-term. These are hard to quantify, but also cannot be ignored. **First**, there is real commoditization risk of the inference CDU ecosystem. Brazos specs. are meaningfully easier to deliver than Deschutes, and we believe they don't really need the engineering premium of a Vertiv or nVent. While these players may still opt to build and service Brazos-specified units, margins will be lower. We think the service attach will still be meaningful, but hypothesize less "moaty" than they would be with training GW where the cost of failure is meaningfully higher. The deciding factor here is inference rack densities; we are unsure what this looks like for purpose-built inference silicon from hyperscaler. If it remains below 60 kW/rack, then a meaningful share could be cooled by L2A Brazos-style CDUs. If not, larger Deschutes-class CDUs will be needed instead (which we would expect have higher margin and stickier service attach). **Second**, we may also see some hyperscaler demand shift from greenfield to retrofits, especially if projects get delayed. While we do not see that in our stranded capacity numbers in our [data center capacity tracker](#), there has been some anecdotal input from companies that they are seeing customers push out deliveries because of a lack of readiness / project delays. With that said, **we do not view this announcement by Google as imminently alarming for CDU manufacturers**, especially those that focus on technical excellence at the top of the stack. Some level of CDU commoditization was always expected by the market. While we think this announcement accelerates the commoditization trend for lower spec. products, we still believe an innovation premium exists for flagship CDUs, and expect it to continue as top-tier liquid cooling technology evolves.

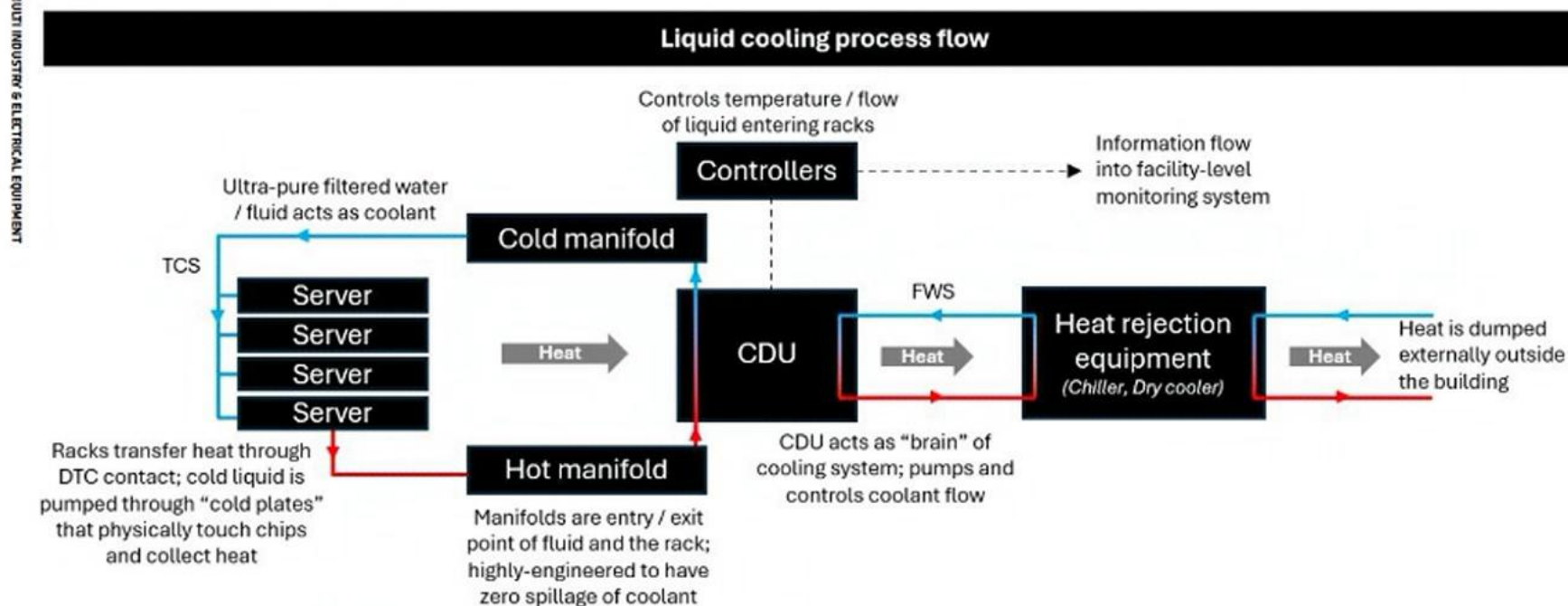
EXHIBIT 4: McKinsey and Co. Estimates of GW add by category

Global data center demand by workload, 2025–30, gigawatts



Note: Includes all provider types.
¹Per annum.
 Source: McKinsey Data Center Demand Model

Source: Company Reports (McKinsey and Co.)

EXHIBIT 5: **Recap: How does Liquid Cooling work?****Legend**

Cold liquid

Hot liquid

Flow of data

Glossary of key liquid cooling equipment

Cold Plates



Coolant Distribution Unit (CDU)



Manifold

Glossary of liquid cooling terms

- DTC (Direct to chip)
- TCS (Technology cooling system)
- FWS (Facility water system)
- TDP (Thermal Design Power)

Source: Bernstein Analysis

EXHIBIT 6: Recap: Our CDU outlook

CDUs | We think CDUs are here to stay; some commoditization risk (but less than cold plates)

Timeframe	Near-term (2026 – 2028)	Mid-term (2028 – 2030)	Long-term (2030+)
Risk of Commoditization	Low	Moderate	Moderate
Risk of Obsolescence	Low	Low	Low
Commentary	Near-term, we expect CDUs to be a growth driver for anyone with exposure to liquid cooling from DCs We expect the market to be supply constrained; anyone with a CDU offering will probably see large backlogs through 2028 While there are some commoditization risks from programs like OCP, Project Deschutes, etc. we believe the pace of product evolution largely supports vendor pricing power	Regardless of whether immersion cooling or DTC is the dominant configuration at this point, CDUs will still be required (so there is no obsolescence risk) We do expect the liquid cooling market to be much more mature at this point and think there is larger possibility that specs. become standardized However, service quality / uptime will remain relevant (especially if CDUs move to the facility level vs. rack level), enabling CDU players to drive revenue from their install base vs. new equipment sales	Even in a config. with silicon etched microchannels, CDUs will remain relevant (and if anything, become more important as the flow volume / pressure become critical to finely control) We think CDUs will be able to retain some pricing power long-term even if they are fully specified by hyperscalers at this point – driven by a combination of higher complexity (esp. compared to cold plates) and the need for service once CDUs installed Given the high value of racks, we do not think the service can be outsourced to third-party non-OEMs

Source: Bernstein Analysis and Estimates, Company Reports

Practically no obsolescence risk; we think CDUs are here to stay regardless of the dominant cooling config.

Some commoditization risk; hyperscalers will specify designs but CDUs are more complex (multiple connected pieces) than cold plates which preserves pricing power

In addition, CDUs need service (which cold plates do not) **which also defends margin**

We think CDUs are a great business to be in, ESPECIALLY for companies that can drive the innovation roadmap vs. just become contract mfg.

We see wide dispersion of forecasts for broader CDU market; **but an LSD \$B market size today growing MSD/HSD \$B in the next 5 years does not seem too improbable**

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VALUATION METHODOLOGY**Vertiv Holdings Co**

We value Vertiv at 32x NTM + 1 EBITDA of \$5.1B to reach our target price of \$416.

nVent Electric PLC

We used a SOTP approach to value NVT. For the EC business, we used NTM + 1 EBITDA of \$417M and an EV/EBITDA multiple of 17x. For the SP business, we use an NTM + 1 EBITDA of \$1,063M and an EV/EBITDA multiple of 28x. Based on these assumptions, we arrive at a fair value per share of \$218.

Schneider Electric SA

Our PT of €310 for Schneider is based on DCF with a WACC of 7.8% and a terminal growth rate of 2.5%. We think a DCF is appropriate for Schneider as it's a cash generative company and delivers stable cash flows irrespective of the point in the cycle.

Eaton Corp PLC

We arrive at our \$534 price target by applying a 27x P/E multiple on our 2030 EPS estimate. Discounting this back, we arrive at our target price which implies a 33x P/E on our 2027 EPS. We believe this is appropriate given the long-term secular factors favoring the strength and long-term resilience of ETN's earnings algorithm.

RISKS**Vertiv Holdings Co**

Downside: (i) Efficiency increasing to the point where less cooling needed, (ii) Unexpected slowing of data center expansion, (iii) Faster than expected shift away from NVIDIA chips to custom silicon (impacts Vertiv because of strong relationships), (iv) Tech. shift away from DTC / 800 VDC if Vertiv can't react fast enough

nVent Electric PLC

Downside: (i) Faster than expected commoditization of CDUs / OCP makes them a commodity, (ii) Production line weakness for CDUs (given NVT is ramping), (iii) Longer than expected lead time to hire for key roles (e.g., open supply chain VP position)

Schneider Electric SA

Key downside risks to our price target and forecasts: 1) We assume growth rate of c.12% for 2026 which assumes continued strength in Energy Management and improvement in IA. 2) Pull back in data center demand given technology changes which alter the composition of electrical infrastructure required in data centers and therefore demand for certain Schneider Electric products. 3) Imposition of additional tariffs in the US that lower demand for Schneider's products given weaker consumer buying power (we expect Schneider to be able to pass on any tariff costs, but demand may not be as elastic).

Eaton Corp PLC

Downside risks include: 1) lower than expected load growth impacting ETN's Electrical organic growth business, 2) interruptions to the ongoing global data center build, 3) slower than expected utility capex growth, 4) slower than expected reshoring trends, and 5) slower than expected growth/cancellations in mega projects.

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The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

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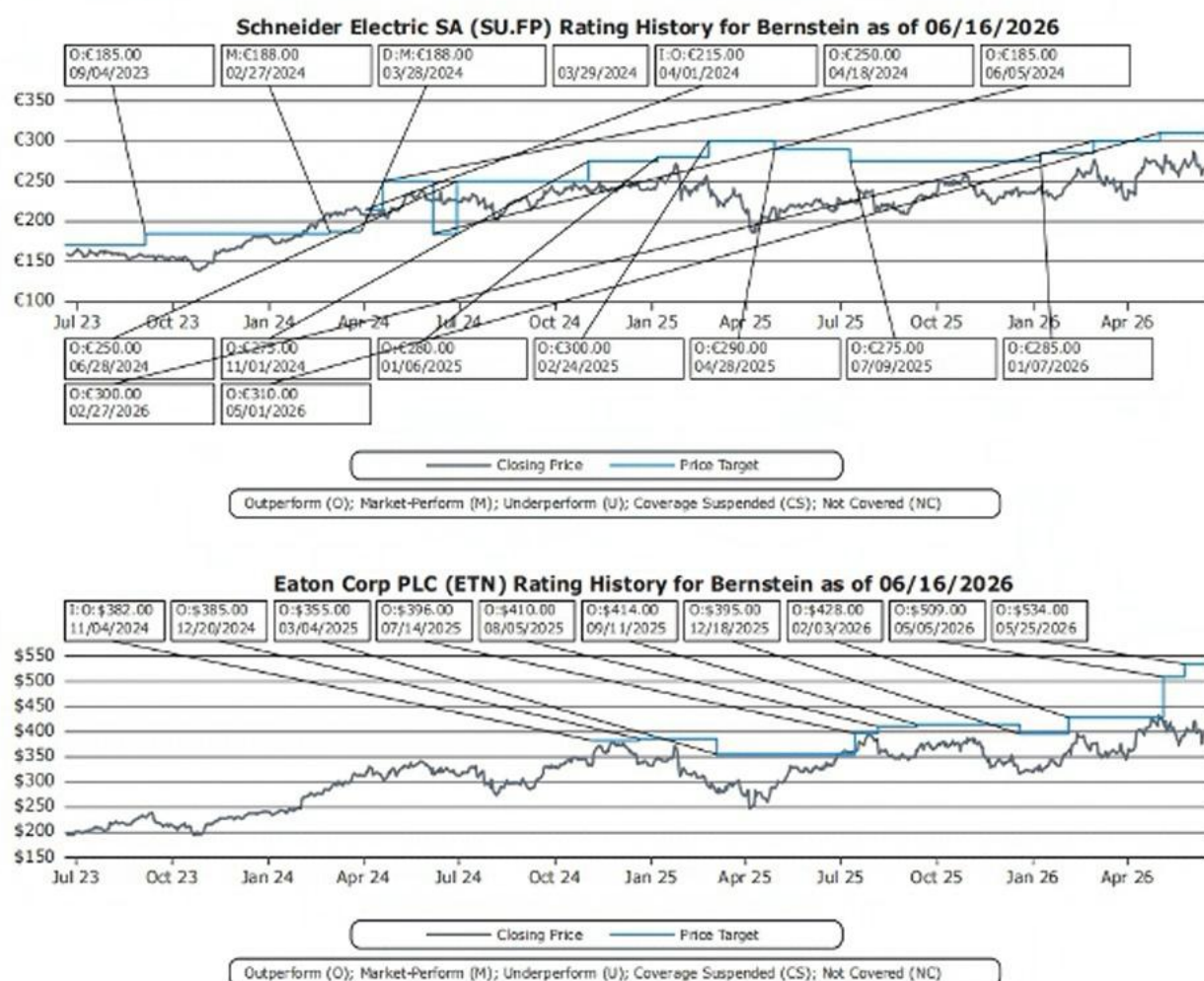
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Outperform	BUY	51.1%	16.5%
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As of March 31, 2026. All figures are updated quarterly.

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All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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